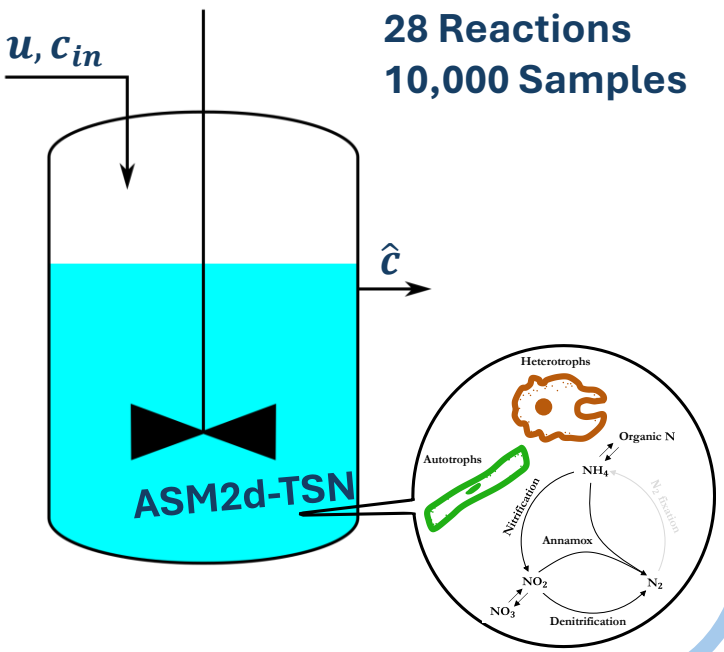
Steady-State CSTR Simulation



20 States
28 Reactions
10,000 Samples

ICSOR Framework

Inputs: $\boldsymbol{u}, \boldsymbol{c_{in}}, \boldsymbol{\phi}(\boldsymbol{u}, \boldsymbol{c_{in}})$

Coupled driver:
$\boldsymbol{d} = \boldsymbol{B\phi}, \boldsymbol{R} = \boldsymbol{I} - \boldsymbol{\Gamma}$

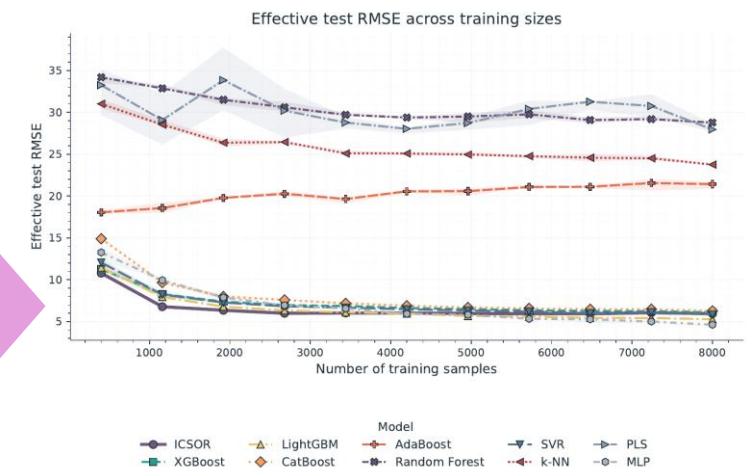
Hierarchical deployment:
Raw → Affine → LP

Physically admissible state $\hat{\boldsymbol{c}}$

Mass Conservation
$\boldsymbol{Ac} = \boldsymbol{Ac_{in}}$

Non-Negativity
$\boldsymbol{c} \geq \boldsymbol{0}$

Benchmark Comparison



0% Mass Violations
Non-Negativity Violations

Interpretable

Accurate